XENIA’26

HACKATHON

BUILD. INNOVATE. COMPETE

- **Problem Statement ID :** NMA1
- **Problem Statement Title :** Scalable Learning Platform for High Traffic Events
- **PS Domain :** NeuraMach.AI
- **Team Name :** POOKIE PANDAS

Proposed Solution

GITHUB LINK DEMO LINK

Problem : Organizations often struggle with securely hosting and delivering large files (videos, PDFs) while keeping costs low and preventing unauthorized access.

Content Delivery using CDN

Videos and static content are delivered through a CDN from the nearest location, reducing backend load and ensuring fast streaming for many users.

Contest/Test Service for Exams

Exam questions are cached in Redis, submissions are placed in a queue, and worker servers process them while automatically scaling during high load.

API Gateway Controls Requests

All API requests pass through the API Gateway, which handles authentication, rate limiting, and routing so only valid requests reach backend services.

Authentication & User Management

Manages login, user identity, and access permissions to ensure secure platform access.

Load Balancer Distributes Traffic

Incoming traffic is distributed evenly across servers, preventing overload on any single server.

Auto-Scaling Infrastructure

Servers automatically scale up during peak traffic and scale down during low usage to maintain performance and reduce costs.

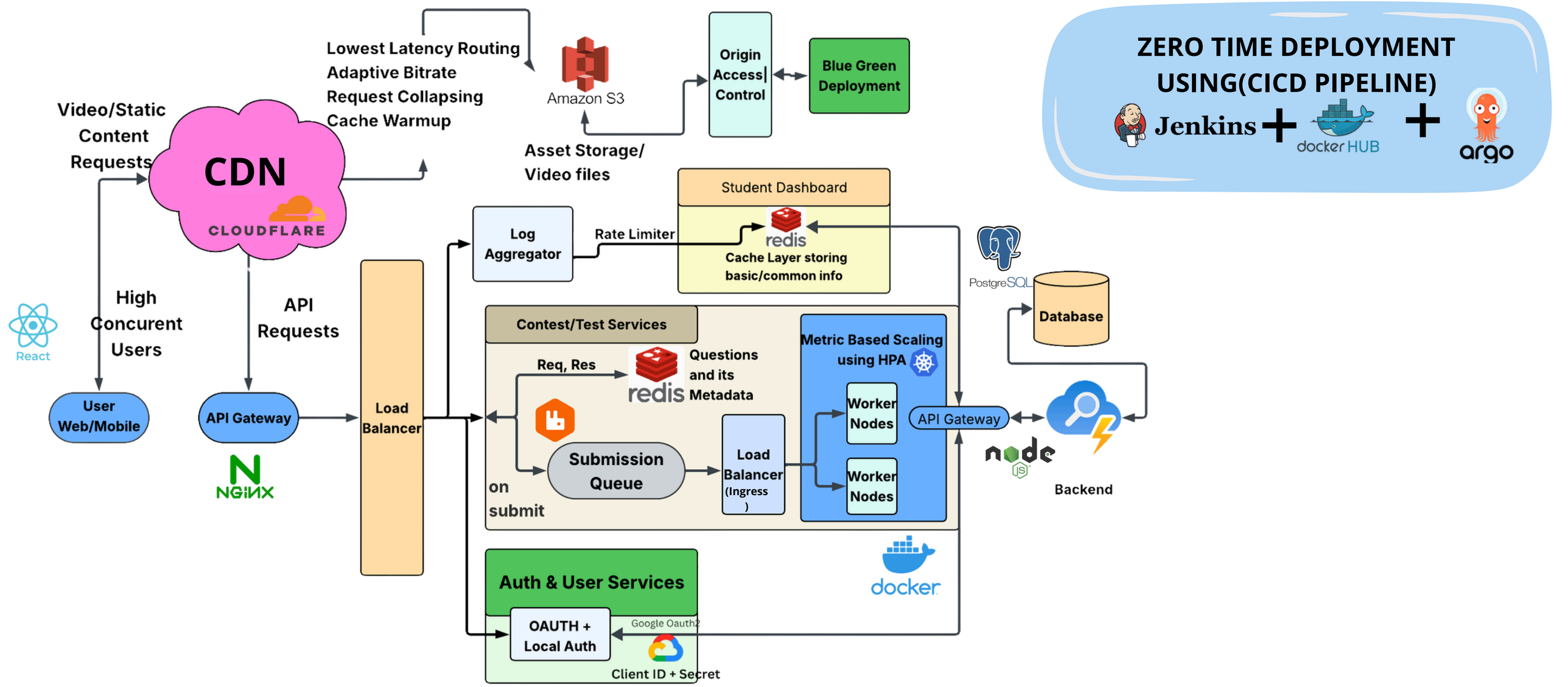
Student Dashboard Service

Provides course access, progress tracking, results, and notifications, using Redis caching to reduce database load and improve response speed.

Zero Downtime Deployment

Blue-Green deployment allows system updates without interrupting users, ensuring continuous availability.

SYSTEM ARCHITECTURE AND TECH STACK



FEASIBILITY & VIABILITY

Technical Feasibility

- Uses proven technologies: CDN, API Gateway, Load Balancer, Redis, Auto-Scaling.
- Microservices architecture ensures modular development and easier maintenance.
- Blue-Green deployment enables zero downtime updates.
- Redis caching reduces database load and improves performance.

Economic Viability

- Pay-as-you-use cloud model reduces upfront cost.
- Auto-scaling minimizes infrastructure cost during low traffic.
- CDN reduces backend server expenses.
- Long-term cost savings through optimized resource utilization.

Operational Feasibility

- Separate services reduce system complexity.
- Easy monitoring and scaling during peak exam traffic.
- Automated scaling reduces manual intervention.

Performance Viability

- CDN reduces latency for global users.
- Redis caching minimizes database response time.
- Load balancer ensures no single server becomes a bottleneck.

IMPACT & BENEFITS

For Students

- Fast content delivery with CDN.
- Smooth exam experience even during high traffic.
- Real-time results and progress tracking.
- Secure authentication and data privacy.

For Institution

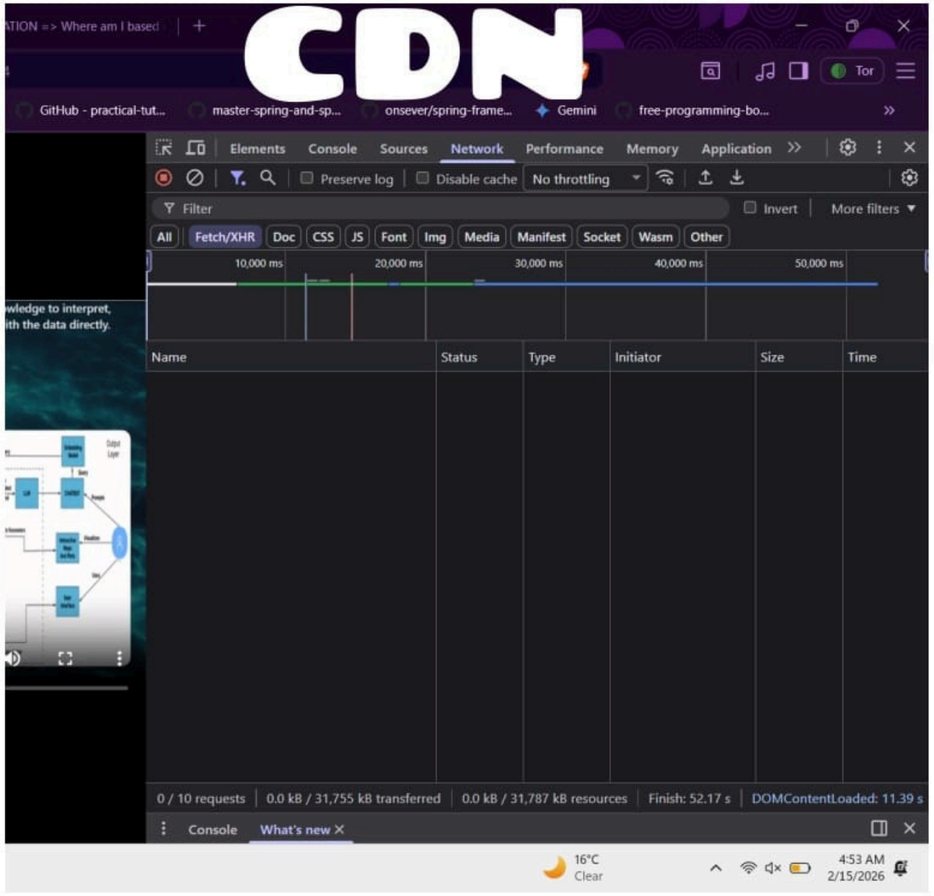
- Handles large concurrent users without crashes.
- Zero downtime during updates.
- Reduced operational cost due to auto-scaling.
- Reliable and secure exam management system.

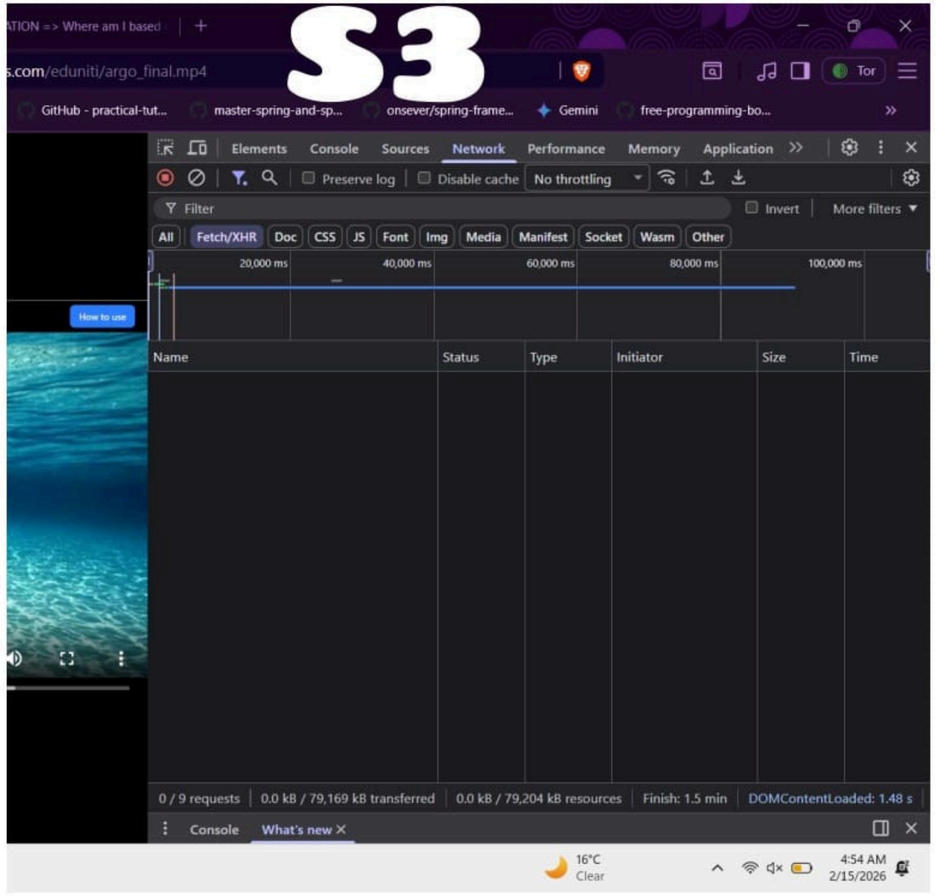
System-Level Benefits

- High Availability Architecture ensures continuous service even if one server fails.
- Fault Tolerance prevents complete system breakdown during partial failures.
- Horizontal Scalability supports unlimited user growth without redesign.
- Low Latency Performance through CDN and caching mechanisms.
- Distributed Processing improves exam submission handling under heavy load.
- Resource Optimization via auto-scaling reduces idle infrastructure cost.
- Centralized API Management improves monitoring, logging, and security control.

RESEARCH AND REFERENCES

[1] M. Kleppmann, Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems. Sebastopol, CA, USA: O'Reilly Media, 2017.
[2] S. Newman, Building Microservices: Designing Fine-Grained Systems, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2021.
[3] Amazon Web Services, "Auto Scaling Documentation," 2024. [Online]. Available: https://docs.aws.amazon.com
[4] Microsoft Azure, "High Availability and Deployment Strategies," 2024. [Online]. Available: https://learn.microsoft.com/azure
[5] Google Cloud Platform, "Cloud Load Balancing and CDN Overview," 2024. [Online]. Available: https://cloud.google.com/docs
[6] Redis Labs, "Redis Documentation," 2024. [Online]. Available: https://redis.io/docs
[7] Cloud Native Computing Foundation, "Cloud-Native and Microservices Best Practices," 2024. [Online].





LOCATION	STATUS	DNS	CONNECT	TLS	TTFB
Frankfurt	200	26.41 ms	0.68 ms	16.2 ms	20.5 ms
Amsterdam	200	31.38 ms	2.06 ms	17.43 ms	23.27 ms
London	200	15.07 ms	2.68 ms	16.42 ms	21.8 ms
New York	200	24.72 ms	3.98 ms	23.57 ms	39.52 ms
San Francisco	200	29.41 ms	1.83 ms	15.15 ms	22.11 ms
Singapore	200	13.25 ms	1.83 ms	12.59 ms	18.07 ms
Sydney	200	17.48 ms	1.31 ms	101.86 ms	107.56 ms

Frankfurt	200	21.14 ms	128.46 ms	145.02 ms	425.43 ms
Amsterdam	200	16.22 ms	146.45 ms	160.07 ms	477.41 ms
London	200	11.46 ms	142.21 ms	151.97 ms	452.99 ms
New York	200	17.56 ms	203.26 ms	214.38 ms	640.69 ms
San Francisco	200	17.21 ms	265.16 ms	275.75 ms	828.44 ms
Singapore	200	10.02 ms	60.89 ms	72.1 ms	214.15 ms
Sydney	200	10.75 ms	154.44 ms	244.34 ms	568.77 ms
Bangalore	200	24.46 ms	15.49 ms	24.77 ms	75.04 ms

```
NAME                      READY   STATUS    RESTARTS      AGE
api-779588fc69-6fxtl      1/1     Running   0             3h13m
api-779588fc69-8kt4t      1/1     Running   0             3h13m
rabbitmq-6b765cd6ff-s4fnm 1/1     Running   0             3h13m
worker-5cc7979d44-j72hd   1/1     Running   1 (14m ago)   16m
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl top pods
NAME                      CPU(cores)   MEMORY(bytes)
api-779588fc69-6fxtl      1m           20Mi
api-779588fc69-8kt4t      1m           20Mi
rabbitmq-6b765cd6ff-s4fnm 4m           155Mi
worker-5cc7979d44-j72hd   1m           12Mi
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl top nodes
NAME       CPU(cores)   CPU%   MEMORY(bytes)   MEMORY%
minikube   222m         1%     1131Mi          4%
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl get hpa -w
NAME         REFERENCE           TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
worker-hpa   Deployment/worker   cpu: 0%/60%   1         8         1          3h14m
```

```
    -H "Content-Type: application/json" \
    -d '{"data":"test"}' \
    http://192.168.49.2/submit

Summary:
  Total:        31.5199 secs
  Slowest:      1.8898 secs
  Fastest:      0.0008 secs
  Average:      0.3057 secs
  Requests/sec: 634.5191

  Total data:   9032519 bytes
  Size/request: 451 bytes

Response time histogram:
  0.001 [1]     |
  0.190 [7886]  |■■■
```

```
Listing queues for vhost / ...
name    messages
submit_queue    9501
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl get hpa -w
NAME         REFERENCE           TARGETS        MINPODS   MAXPODS   REPLICAS   AGE
worker-hpa   Deployment/worker   cpu: 92%/60%   1         8         2          3h46m
^C(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl top nodes
NAME       CPU(cores)   CPU%   MEMORY(bytes)   MEMORY%
minikube   953m         7%     1341Mi          5%
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl top pods
NAME                          CPU(cores)   MEMORY(bytes)
api-59944cccfd-l8gnh          103m         79Mi
api-59944cccfd-mqft2          106m         69Mi
rabbitmq-5d7f756df4-8j2zl     93m          161Mi
worker-6c7f64f574-4c62h       184m         15Mi
worker-6c7f64f574-8c5mt       184m         15Mi
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl exec -it rabbitmq-5d7f756df4-8j2zl -- rabbitmqctl list_queues
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages
submit_queue    2390
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl get pods
NAME                          READY   STATUS    RESTARTS        AGE
api-59944cccfd-l8gnh          1/1     Running   0               5m53s
api-59944cccfd-mqft2          1/1     Running   0               5m52s
rabbitmq-5d7f756df4-8j2zl     1/1     Running   0               5m53s
```

```
api-59944cccfd-l8gnh          1/1     Running   0               5m53s
api-59944cccfd-mqft2          1/1     Running   0               5m52s
rabbitmq-5d7f756df4-8j2zl     1/1     Running   0               5m53s
worker-6c7f64f574-4c62h       1/1     Running   1 (5m16s ago)   5m53s
worker-6c7f64f574-8c5mt       1/1     Running   0               5m52s
worker-6c7f64f574-f4nrv       1/1     Running   0               69s
worker-6c7f64f574-pxq45       1/1     Running   0               2m10s
worker-6c7f64f574-qqgp6       1/1     Running   0               2m10s
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl exec -it rabbitmq-5d7f756df4-8j2zl -- rabbitmqctl list_queues
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages
submit_queue    0
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl get pods
NAME                          READY   STATUS    RESTARTS        AGE
api-59944cccfd-l8gnh          1/1     Running   0               9m48s
api-59944cccfd-mqft2          1/1     Running   0               9m47s
rabbitmq-5d7f756df4-8j2zl     1/1     Running   0               9m48s
worker-6c7f64f574-4c62h       1/1     Running   1 (9m11s ago)   9m48s
worker-6c7f64f574-8c5mt       1/1     Running   0               9m47s
worker-6c7f64f574-f4nrv       1/1     Running   0               5m4s
worker-6c7f64f574-pxq45       1/1     Running   0               6m5s
worker-6c7f64f574-qqgp6       1/1     Running   0               6m5s
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl get hpa -w
NAME         REFERENCE           TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
worker-hpa   Deployment/worker   cpu: 0%/60%   1         8         5          3h53m
```

```
t@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl get hpa -w
  REFERENCE              TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
  Deployment/worker      cpu: 123%/60%    1         8         5          16m
  Deployment/worker      cpu: 123%/60%    1         8         8          16m
Bworker-hpa   Deployment/worker   cpu: 15%/60%     1         8          8
  Deployment/worker      cpu: 0%/60%      1         8         8          18m
  Deployment/worker      cpu: 0%/60%      1         8         8          22m
  Deployment/worker      cpu: 0%/60%      1         8         2          22m
  Deployment/worker      cpu: 0%/60%      1         8         2          23m
  Deployment/worker      cpu: 0%/60%      1         8         1          23m
  Deployment/worker      cpu: 0%/60%      1         8         1          67m
  Deployment/worker      cpu: 0%/60%      1         8         1          67m
```

```
NAME                      READY   STATUS    RESTARTS      AGE
api-779588fc69-6fxtl      1/1     Running   0             3h13m
api-779588fc69-8kt4t      1/1     Running   0             3h13m
rabbitmq-6b765cd6ff-s4fnm 1/1     Running   0             3h13m
worker-5cc7979d44-j72hd   1/1     Running   1 (14m ago)   16m
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl top pods
NAME                      CPU(cores)   MEMORY(bytes)
api-779588fc69-6fxtl      1m           20Mi
api-779588fc69-8kt4t      1m           20Mi
rabbitmq-6b765cd6ff-s4fnm 4m           155Mi
worker-5cc7979d44-j72hd   1m           12Mi
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl top nodes
NAME       CPU(cores)   CPU%   MEMORY(bytes)   MEMORY%
minikube   222m         1%     1131Mi          4%
(base) rohit@rohit-Nitro-AN515-44:~/Desktop/eduniti/worker$ kubectl get hpa -w
NAME         REFERENCE           TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
worker-hpa   Deployment/worker   cpu: 0%/60%   1         8         1          3h14m
```